Course Code Course Title: EP lab

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	3	2	2	3	2	2	3	3	3	3	3	2	3
CO2	2	3	3	2	3	0	2	3	3	3	3	3	2	3
EP lab	2	3	3	2	3	2	2	3	3	3	3	3	2	3

CO1: Learn to assemble and disassemble the PC, monitor CPU performance and troubleshoot various problems related to it.

CO2 Implement the electronic circuits using electronic components and Equipments.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
co1	3	3	3	0	3	0	0		3	3	3	3	0	3
co2	3	3	3	0	3	0	0		3	3	3	3	0	3
co-po	3	3	3	0	3	0	0	3	3	3	3	0	3	

	PO 1: Engineering knowledge:				CO1	CO2
	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization for the solution of complex engineering					
	Competency		Indicators			
	1.1	Demonstrate competence in mathematical modeling	1.1.1	Understand concepts of linear algebra, calculus, transforms differential equations, discrete mathematics, graph theory, probability, statistics needed to solve Computing problems		
			1.1.2	Apply concepts of linear algebra, calculus, transforms differential equations, discrete mathematics, graph theory, probability, statistics needed to solve Computing problems		
	1.2	Demonstrate competence in basic sciences	1.2.1	Understand basic sciences concepts (physics, chemistry and social sciences) needed for solving computing problems		
			1.2.2	Apply basic sciences concepts (physics, chemistry and social sciences)to solve computing problems	y	y
	1.3	Demonstrate competence in core engineering fundamentals	1.3.1	Understand core engineering concepts needed in computing problems	y	y
			1.3.2	Apply core engineering concepts to solve computing problems	y	y

	1.4	Demonstrate competence in specialized engineering knowledge to the program(CSE)	1.4.1	Apply basic computing to sove real time computing problems	y	y
	(CSE)				7	7
	Mapped Performance Indicator (MPI)				4	4
	Mapping Percentage				14285	14285
	Mapping Level				2	2

PO 2: Problem analysis				CO1	CO2
Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.					
Competency		Indicators			
2.1	Demonstrate an ability to identify and formulate complex engineering problem	2.1.1	Problem Identification	y	y
		2.1.2	Formulating objectives	y	y
2.2	Demonstrate an ability to to perform literature survey and conclude on gaps	2.2.1	Perform literature survey		
		2.2.2	Identify gaps		
2.3	Demonstrate ability to analyse existing solutions and propose analysis models	2.3.1	Analyse existing solutions	y	y
		2.3.2	Propose analysis models for the identified problem	y	y
2.4	Demonstrate an ability to reach conclusions using principles of mathematics, physics, chemistry, social sciences, core engineering and computer science	2.4.1	Identify gaps and conclude	y	y
			Total Indicators	7	7
			Mapped Performance Indicator (MPI)	5	5
			Mapping Percentage	71.4	71.4
			Mapping Level	3	3

					CO1	CO2
	PO 3: Design/Development of Solutions:					
	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.					
	Competency		Indicators			
	3.1	Ability to analyse Design solutions for complex engineering problems	3.1.1	Able to define a precise problem statement with objectives and scope.	y	y
			3.1.2	Elicit and document, engineering design requirements from stakeholders		y
			3.1.3	Able to review the state of art literature	y	y
			3.1.4	Able to select appropriate quality attributes from SE/ hardware standards	y	y
			3.1.5	Explore and synthesize engineering system requirements from health, safety risks, environmental, cultural and societal concerns	y	y
			3.1.6	Determine design objectives and functional requirements to develop product specifications from concepts	y	y
	3.2	Ability to suggest design solutions for components / processes o meet the specified needs	3.2.1	Ability to apply formal idea generation tools to develop multiple / alternate design solutions		
			3.2.2	Ability to build models/prototypes & to develop a diverse set of design solutions		y
			3.2.3	Identify suitable criteria for the evaluation of alternate design solutions		
	3.3	Refine and evaluate design solutions	3.3.1	Ability to select optimal design solutions for further development		y
			3.3.2	Consult with domain experts and stakeholders to select candidate engineering design solution for further development		
			3.4.1	Refine a conceptual design into a detailed design with the existing constraints		y
			3.4.2	Generate information through appropriate tests to improve or revise the design	y	y

	Total Indicators	13	13
	Mapped Performance Indicator (MPI)	6	10
	Mapping Percentage	.153846	.923076
	Mapping Level	2	3

	PO 4: Conduct investigations of complex problems:				CO1	CO2
	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.					
	Competency		Indicators			
	4.1	Demonstrate an ability to conduct investigations of complex engg problems	4.1.1	Define a problem, its scope, objectives and importance for investigation purposes	y	y
			4.1.2	Examine/Identify the relevant methods, tools and techniques for experimentation and data analysis	y	y
	4.2	Demonstrate an ability to design experiments to solve open-ended problems	4.2.1	Design and develop an experimental approach using tools for the objectives	y	y
	4.3	Demonstrate an ability to analyze data and reach a valid conclusion	4.3.1	Analyze critically the data / results for trends and correlations, stating possible errors and limitations		
			4.3.2	Represent data / results (in tabular and/or graphical forms) so as to facilitate analysis and explanation of the data, and drawing of conclusions		
			4.3.3	Synthesize information and knowledge to validate conclusions		
	Total Indicators				6	6
	Mapped Performance Indicator (MPI)				3	3
	Mapping Percentage				50	50
	Mapping Level				2	2

	PO 5: Modern tool usage:				CO1	CO2
	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.					
	Competency		Indicators			
	5.1	Demonstrate an ability to identify/create modern engineering tools, techniques and resources for solving engg problems	5.1.1	Identify modern engineering tools such modeling and analysis tools, techniques and resources	y	y
			5.1.2	Create/adapt/modify/extend tools and techniques to solve enggproblems		
	5.2	Demonstrate an ability to select and apply computer science specific tools, techniques and resources	5.2.1	Identify the strengths and limitations of tools for (i) acquiring information, (ii) modeling and simulating, (iii) monitoring system performance, and (iv) analysis, design, coding and testing		
			5.2.2	Apply suitable tools and techniques to solve Computer Science problems	y	y
	5.3	Demonstrate an ability to evaluate the suitability and limitations of tools used to solve an engineering problem	5.3.1	Discuss limitations and validate tools, techniques and resources	y	y
			5.3.2	Verify the credibility of results from tool use with reference to the accuracy and limitations, and constraints	y	y
	Total Indicators				6	6
	Mapped Performance Indicator (MPI)				4	4
	Mapping Percentage				66666	66666
	Mapping Level				3	3

	PO 6: The Engineer and Society:				CO1	CO2
	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent					
	Competency		Indicators			
	6.1	Demonstrate an ability to describe engineering roles in a broader context, e.g. pertaining to the environment, health, safety, legal and public welfare	6.1.1	Identify and describe various engineering roles and its impact on societal, health, safety, legal & ethical issues		
	6.2	Demonstrate an understanding of professional engineering regulations, legislation and standards	6.2.1	Interpret codes, and standards, rules and regulations relevant to Computing Industry in relational to societal, legal, safety, health and cultural issues	y	
	Total Indicators				2	2
	Mapped Performance Indicator (MPI)				1	0
	Mapping Percentage				50	0
	Mapping Level				2	0

					CO1	CO2
PO 7: Environment and sustainability:						
Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and the need for sustainable development.						
Competency		Indicators				
7.1	Demonstrate an understanding of the impact of engineering and industrial practices on social, environmental issues in economic contexts	7.1.1	Identify risks/impacts in the life-cycle of an hardware/software product / process		y	
		7.1.2	Understand the relationship between the technical aspects and social/environmental/economic factors for sustainability			y
7.2	Demonstrate an ability to apply principles of sustainable design and development	7.2.1	Describe various approaches to support sustainable development of hardware / software		y	y
		7.2.2	Apply professional engineering practices to development hardware / software considering social and environmental contexts			
				Total Indicators	4	4
				Mapped Performance Indicator (MPI)	2	2
				Mapping Percentage	50	50
				Mapping Level	2	2

	PO 8: Ethics:				CO1	CO2
	Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.					
	Competency		Indicators			
	8.1	Demonstrate an ability to recognize ethical dilemmas	8.1.1	Understand various ethical practices in profession, and responsibilities /regulations in computing	y	y
	8.2	Demonstrate an ability to apply the Code of Ethics	8.2.1	Apply professional code of ethics / ethical practices	y	y
	Total Indicators				2	2
	Mapped Performance Indicator (MPI)				2	2
	Mapping Percentage				100	100
	Mapping Level				3	3

PO 9: Individual and team work:					CO1	CO2
Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.						
Competency		Indicators				
9.1	Demonstrate an ability to form a team and define a role for each member	9.1.1	team composition & diversity / rubrics			
		9.1.2	Define individual members contribution for effective team work, to accomplish a goal.		y	y
9.2	Demonstrate effective individual and team contributions	9.2.1	individual contribution to problem solving		y	y
		9.2.2	effective team communication		y	y
		9.2.3	ability to resove conflicts		y	y
		9.2.4	demonstrate leadership skills and integrity			
9.3	Demonstrate success in a team-based project	9.3.1	Present results as a team, with smooth integration of contributions from the individual team members		y	y
				Total Indicators	7	7
				Mapped Performance Indicator (MPI)	5	5
				Mapping Percentage	42857	42857
				Mapping Level	3	3

PO 10: Communication:

Communicate effectively on complex engineering activities with the engineering community and with the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions

Competency		Indicators		CO1	CO2
10.1	Demonstrate an ability to comprehend technical literature and document the study	10.1.1	Read, understand and interpret technical and non-technical information	y	y
		10.1.2	Produce clear, well-constructed, and well-supported literature survey/ documentation (analysis / design)		
		10.1.3	Organise a document or presentation - a logical progression of ideas with clarity	y	y
10.2	Demonstrate competence in listening, speaking, and presentation	10.2.1	Listen to and comprehend information, instructions, and viewpoints of others	y	y
		10.2.2	Deliver effective oral presentations to technical and non- technical audiences	y	y
10.3	Demonstrate the ability to integrate different modes of communication	10.3.1	Create standard models / drawings to complement writing and presentations	y	y
		10.3.2	Use a variety of media effectively to convey a message in a document or a presentation	y	y
			Total Indicators	7	7
			Mapped Performance Indicator (MPI)	6	6
			Mapping Percentage	71428	71428

	Mapping Level	3	3

PO 11: Project management and finance:				CO1	CO2	
Demonstrate knowledge and understanding of the engineering and management principles and apply these to one’s work, as a member and leader in a team, to manage projects and in multidisciplinary environments.						
Competency		Indicators				
11.1	Understand engineering / management principles for SE/hardware projects	11.1.1	Understand various approaches / hardware techniques for engineering software	y	y	Describ e various econom
		11.1.2	Understand various approaches / techniques for managing software / hardware processes and products	y	y	Analyze different forms of financial
11.2	Demonstrate an ability to compare and contrast the costs/benefits of alternate approches used	11.2.1	Analyze and select the most appropriate and effective technique based on economic and financial considerations.	y	y	
11.3	Demonstrate an ability to plan/manage an engineering activity within time and budget constraints	11.3.1	Identify the tasks and the resources required to complete/ manage the process / product	y	y	
		11.3.2	Apply tools / techniques complete / manage the process / product on time and on budget			
			Total Indicators	5	5	
			Mapped Performance Indicator (MPI)	4	4	
			Mapping Percentage	80	80	
			Mapping Level	3	3	

	PO 12: Life-long learning:				CO1	CO2
	Recognise the need for, and have the preparation and ability to engage in independent and life- long learning in the broadest context of technological change.					
	Competency		Indicators			
	12.1	Demonstrate an ability to identify gaps in knowledge and a strategy to close these gaps	12.1.1	Understand the need for continuous professional development	y	y
			12.1.2	Identify deficiencies or gaps in knowledge and demonstrate an ability to source information to close this gap	y	y
	12.2	Demonstrate an ability to identify changing trends in computer engineering knowledge and practice	12.2.1	Ability to understand the changing landscape in Computer Science and Engineering from valid sources	y	y
			12.2.2	Updating knowledge to apply state ofart tools and techniques in their profession		
	Total Indicators				4	4
	Mapped Performance Indicator (MPI)				3	3
	Mapping Percentage				75	75
	Mapping Level				3	3

					CO1	CO2
	PSO1					
	Conceptualize domain specific problems, analyze and use algorithms, tools and techniques to provide solutions					
	Competency		Indicators			
	1.1	Demonstrate competence in identifying domain specific problem	1.1.1	Identify domain specific problems	y	y
			1.1.2	identify basic sciences, core engineeing, computing concepts to be applied	y	y
	1.2	Demonstrate the capability to analyse the problem	1.2.1	Analyse the problem using suitable tools and techniques	y	y
			1.2.2	Able to conclude on analysis and move to design solutions	y	y
	1.3	Apply suitable algorithms to solve the problems	1.3.1	Able to suggest approaches / algorithms to solve the problem		
			1.3.2	Evaluate and suggest efficient/ effective approach		
	1.4	Apply SE / computing tools	1.4.1	Apply tools for problem solving		
	Total Indicators				7	7
	Mapped Performance Indicator (MPI)				4	4
	Mapping Percentage				.142857	.142857
	Mapping Level				2	2

						CO1	CO2
	PSO2						
	Apply standard / best industry practices and techniques in developing Software / Hardware Systems						
	Competency		Indicators				
	1.1	Understand Industrial and professional practices	1.1.1	Identify and comprehend practices and standards in software/hardware		y	y
			1.1.2	suggest standards for the given problem		y	y
	1.2	Comprehend Industrial practices for Analysing the problem	1.2.1	Identify and comprehend analysis practices and models		y	y
			1.2.2	suggest suitable analysis models for a given problem			
	1.3	Comprehend Industrial practices for designing solutions	1.3.1	Identify and comprehend design practices and models		y	y
			1.3.2	suggest suitable design models for a given problem		y	y
	1.4	Evaluate the system	1.4.1	Suggest suitable tools/practices/techniques to evaluate / test systems			
					Total Indicators	7	7
					Mapped Performance Indicator (MPI)	5	5
					Mapping Percentage	.4285714	.4285714
					Mapping Level	3	3